

**ENERGY SYSTEMS AND ACTUATION TECHNOLOGIES FOR NEXT GENERATION
HUMANOID AND GENERAL ROBOTICS**

(General Environmental Harvesting of Energy in Humanoid and General Robotics)

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CHAPTER ONE

INTRODUCTION

1.1 Introduction To Energy Harvesting from Motion

Humanoid robots are increasingly being developed to perform tasks that require human-like dexterity, mobility, and decision making capabilities. From healthcare assistance to disaster response, the versatility of these robots makes them valuable across a wide range of industries and not just humanoid robots but also general robots. However, a major limitation to their performance and long-term deployment lies in the power systems that drive their motion and onboard electronics. Most humanoid robots rely heavily on batteries, which provide limited operational time and require frequent recharging. This constraint not only affects their autonomy but also restricts their scalability in real world environments.

The increasing autonomy of humanoid and general-purpose robots has consequently increased the importance of their energy systems. A robot requires energy not only for locomotion but also for actuation, sensing, computation, communication, and interaction with its environment. The energy system must therefore provide sufficient energy capacity, power output, reliability, and operating duration while remaining within the mass and dimensional constraints of the robotic platform. The selection of a robot's energy system also affects its mechanical architecture, electronics, packaging, weight, and operating capabilities. Current robotic systems predominantly use electrochemical energy storage, particularly batteries, as their primary onboard energy source. Batteries provide relatively high energy density and can supply energy for extended periods, making them suitable for mobile and autonomous systems. However, battery-powered operation remains constrained by the relationship between energy capacity, mass, charging time, degradation, and the power requirements of the robot.

The energy problem is therefore not limited to increasing the capacity of batteries. It also involves reducing unnecessary energy consumption, recovering energy that would otherwise be dissipated, improving energy conversion, and developing systems capable of obtaining energy from available environmental and mechanical sources.

In order to address this challenge, I am exploring sustainable energy solutions that can enhance the energy efficiency and autonomy of humanoid systems. One promising area is (ENERGY HARVESTING FROM MOTION), the process of converting the robot's own physical movements, such as walking, joint articulation, and limb swings, into usable electrical energy. This approach has the potential to supplement existing power systems and reduce dependency on frequent battery charging, thereby extending the robot's operational lifespan.

1.2 Energy Sources For Robotic Systems

The energy available to a robotic system can be broadly divided into **stored energy** and **harvested energy**. Stored energy is energy that is supplied to the robot before or during operation and retained in an onboard storage system. Batteries, supercapacitors, and fuel cells are examples of technologies used for storing or converting energy for robotic applications. Batteries are generally associated with relatively high energy storage capacity, whereas supercapacitors are characterized by high power density and rapid charge and discharge capability. Fuel cells provide another route for converting chemical energy into electrical energy and have been investigated for applications requiring longer operating durations. ([The Advanced Portfolio](#)) Sourced on 17-10-2025.

Harvested energy, in contrast, is obtained from mechanical movement or from the environment during operation. Potential sources include mechanical motion, vibration, solar radiation, wind, waves, water currents, thermal gradients, and other forms of ambient energy. The suitability of each source depends on the operating environment and energy requirements of the robot.

This distinction is particularly important because robotic platforms operate under substantially different physical conditions. A humanoid robot operates primarily through mechanical interaction with the ground and objects, an aerial robot operates within an atmospheric environment, while an underwater robot operates under water pressure, currents, buoyancy, and other marine conditions. Consequently, the available energy sources and appropriate harvesting mechanisms differ between platforms.

1.3 Energy Storage Technologies in Robotics

Energy storage forms an essential part of robotic energy systems because harvested energy is often intermittent and may not coincide temporally with the robot's energy demand. A robot may generate or harvest energy at one point in its operating cycle while requiring energy at another. An energy storage system can therefore act as an intermediary between energy generation and energy consumption.

1.3.1 Batteries

Batteries are currently the principal energy-storage technology for many mobile robotic systems. Their importance results from their ability to store comparatively large amounts of electrical energy within a relatively compact mass and volume. Lithium-based batteries are particularly important in mobile robotics and electric systems.

For humanoid robots, however, battery requirements are becoming increasingly demanding as robots are expected to perform dynamic locomotion and sustained operation. A

2026 review of next generation batteries for humanoid robotics identifies energy density and battery-pack integration as major considerations as humanoid systems move toward more demanding applications. ([PubMed](#)) Sourced on 03-01-2026.

1.3.2 Supercapacitors

Supercapacitors provide a complementary energy-storage technology. Unlike batteries, which are primarily suited to storing larger quantities of energy over longer periods, supercapacitors can deliver and absorb high power over relatively short periods. Their rapid charge-discharge capability makes them relevant to applications involving regenerative braking and repeated transient power demands. Recent reviews identify applications of supercapacitors in regenerative braking, industrial automation, and robotics, as well as their use in hybrid energy-storage systems. ([ScienceDirect](#)) Sourced on 03-01-2026.

This characteristic makes supercapacitors particularly relevant to motion-based energy harvesting. A robotic joint may generate a short burst of electrical energy during deceleration, and a high-power storage device can potentially absorb that energy before it is subsequently supplied to another actuator or returned to the robot's electrical system.

1.3.3 Hybrid Energy Storage

The limitations of individual storage technologies have led to increasing research into hybrid energy storage systems. A hybrid architecture can combine the high energy storage capability of batteries with the high power capability and rapid charge discharge characteristics of supercapacitors.

The resulting energy pathway can therefore be represented as:

Energy source → energy harvester/converter → power management → storage → actuator/electronics

Rather than treating the battery as the only destination for all energy generated by the system. Integration of harvesting and storage has also been identified as an important research direction because independently designed harvesting and storage systems can suffer from intermittency, insufficient output, and mismatched operating characteristics. ([PubMed](#)) Sourced on 03-01-2026.

1.4 Energy Harvesting in Robotic Systems

Energy harvesting refers to the conversion of available ambient, mechanical, or environmental energy into a usable form of energy, most commonly electrical energy. In robotics, energy harvesting can be used either as a supplementary energy source or as part of a regenerative energy-recovery system.

The concept is particularly relevant to autonomous robots because energy that is normally dissipated during operation may potentially be recovered. For example, when an electric actuator decelerates a robotic joint, the mechanical energy associated with the deceleration can potentially be converted back into electrical energy. Similarly, mechanical vibration, deformation, environmental fluid movement, solar radiation, and thermal gradients can provide sources of harvestable energy.

A review of energy harvesting and storage technologies for robots identifies three broad technological approaches: **direct energy harvesting and conversion, electrochemical energy storage and conversion, and wireless energy transmission**. It further emphasizes that robotic energy systems must be selected according to factors including mass, energy density, power output, efficiency, safety, environmental conditions, lifetime, and cost. ([The Advanced Portfolio](#)) Sourced on 03-01-2026.

1.5 Types of Energy Harvesting Relevant to Robotics

1.5.1 Electromagnetic Energy Harvesting

Electromagnetic harvesting converts mechanical motion into electrical energy through electromagnetic induction. In robotic systems, this principle can be incorporated into motors, generators, joint mechanisms, wheels, or other moving components.

One important application is regenerative operation of electric actuators. When an actuator is being driven mechanically in a direction that causes the motor to act as a generator, mechanical energy can be converted into electrical energy and returned to the electrical system.

This is particularly relevant to robotic locomotion because walking and other dynamic movements contain phases of positive and negative mechanical work. Research on energy-efficient robotic actuation identifies negative joint mechanical work as an opportunity for energy recycling through series elastic and backdrivable actuators with regeneration. ([University of Waterloo](#)) Sourced on 03-01-2026.

1.5.2 Piezoelectric Energy Harvesting

Piezoelectric materials generate electrical charge when mechanically deformed. This makes them suitable for systems experiencing repeated bending, compression, vibration, or impact.

Research on piezoelectric energy harvesting from human motion has investigated different materials, structures, and configurations for converting biomechanical movement into electrical energy. ([ScienceDirect](#)) Sourced on 03-01-2026.

Within robotics, piezoelectric harvesting can therefore be investigated for:

- vibration;
- joint deformation;
- foot-ground interaction;
- structural movement;
- robotic skins;
- wearable robotic systems;
- low-power sensors.

1.5.3 Triboelectric Energy Harvesting

Triboelectric nanogenerators convert mechanical movement into electrical energy through contact electrification and electrostatic induction. They are being investigated particularly for small scale, flexible, wearable, and soft robotic systems. Their application is relevant where the required power is relatively low and where sensing and energy generation can be combined within a single structure.

1.5.4 Mechanical and Elastic Energy Recovery

Mechanical energy does not necessarily have to be converted into electricity immediately. Springs and other elastic mechanisms can temporarily store mechanical energy and return it during a subsequent movement. This approach is particularly relevant to legged and humanoid robotics because locomotion naturally involves alternating phases of energy absorption and energy release.

The use of series elasticity and backdrivability has therefore been investigated as a means of exploiting passive mechanical dynamics and recovering energy associated with negative joint work. ([University of Waterloo](#)) Sourced on 03-01-2026.

1.6 Energy Harvesting Applications Across Robotic Platforms

The source of harvestable energy differs according to the robotic platform. This provides a basis for extending motion based energy harvesting research from humanoid robots to general robotics.

1.6.1 Humanoid and Land-Based Robotics

Humanoid robots generate mechanical energy through:

- walking
- running
- jumping

- braking
- stair descent
- limb movement
- joint deceleration
- body movement
- impact
- vibration
- Manipulation

During these movements, some joints perform positive mechanical work while others can experience negative mechanical work. Negative work represents a potential opportunity for mechanical energy recovery. Therefore, the energy pathway can be represented as:

Mechanical motion → actuator/generator → electrical conversion → storage → subsequent robotic operation. For example: **Walking → joint deceleration → regenerative motor → power electronics → battery/supercapacitor.**

This forms the direct connection between existing regenerative actuation research and the proposed investigation of energy harvesting from robotic motion.

1.7 Energy Harvesting in Aerial Robotics

Aerial robots have a different energy requirement because maintaining flight requires continuous energy expenditure. Battery-powered UAVs are therefore constrained by the energy density of their onboard storage system. Current research on UAV energy systems includes batteries, fuel cells, photovoltaic cells, and hybrid power architectures. ([ScienceDirect](#)) Sourced on 05-01-2026.

Solar energy is one of the principal environmental energy sources investigated for aerial robotics. Photovoltaic cells can be integrated into the surface of suitable UAVs to convert solar radiation into electricity. Excess energy generated during periods of sufficient sunlight can be stored in batteries for use when solar input is reduced or unavailable. ([MDPI](#)) Sourced on 05-01-2026.

Mechanical energy harvesting has also been investigated in aerial systems. Research has examined wind induced structural vibration, flapping-wing motion, and dynamic soaring as possible mechanisms for obtaining energy associated with aerial motion and atmospheric conditions. ([ScienceDirect](#)) Sourced on 05-01-2026.

Recent research is increasingly treating the UAV as an integrated energy system involving: **solar harvesting → power conversion → energy storage → propulsion → energy management → flight control.**

A 2026 review of solar-powered UAV systems identifies photovoltaic technology, maximum power point tracking, energy storage, battery management, and energy aware flight control as interconnected components of long-endurance aerial systems. ([ScienceDirect](#)) Sourced on 05-01-2026.

1.8 Energy Harvesting in Underwater Robotics

Underwater robotic systems face different energy constraints because access to conventional energy sources is limited by the surrounding environment. Autonomous underwater vehicles, underwater gliders, profiling floats, and underwater sensor networks therefore require energy systems capable of supporting long-duration operation. Potential environmental energy sources include:

- ocean waves;
- water currents;
- pressure variations;
- thermal gradients;
- salinity gradients;
- underwater solar radiation;
- vibration and mechanical movement.

A 2025 systematic review of ocean-energy harvesting for underwater robotic systems identifies environmental energy harvesting as one of the approaches being investigated for powering underwater devices and networks. The review considers waves, currents and other environmental sources while noting that the suitability of each approach depends strongly on the deployment environment. ([MDPI](#)) Sourced on 05-01-2026.

Wave energy harvesting can use the movement of a marine platform or an integrated mechanism to produce relative motion. Proposed mechanisms include hydrofoils, floating bodies, turbines, pendulums, spring-mass systems, and gyroscopic mechanisms. These mechanical systems can then be coupled with electromagnetic, piezoelectric, triboelectric, or other transducers to generate electrical energy. ([ScienceDirect](#)) Sourced on 05-01-2026.

Thermal gradient energy is another area of research. Temperature differences within the ocean can potentially be used to produce energy for long duration underwater vehicles. However, thermal-gradient and salinity-gradient systems remain largely at the research and development stage, with limitations associated with conversion efficiency, power output, and system integration. ([IDEAS/RePEc](#)) Sourced on 05-01-2026.

Recent research has also investigated hybrid thermal gradient systems for uncrewed underwater vehicles. A 2025 study of a phase change material based system reported hydraulic to electrical conversion efficiency of 58.9% for the studied system while noting that existing

thermal gradient approaches still face limitations in the amount of energy generated per operating cycle. ([ScienceDirect](#)) Sourced on 05-01-2026.

1.9 Current Development of Robotic Energy Systems

The current development of robotic energy systems is therefore moving beyond the use of a single onboard battery toward combinations of **energy storage, energy harvesting, energy conversion, regenerative operation, and energy management**.

For humanoid robots, current research is concerned with increasing battery capacity and improving electric actuation while investigating ways to reduce energy consumption and recover mechanical energy.

For aerial robots, photovoltaic systems and hybrid energy architectures are being investigated to extend flight endurance. Current research increasingly combines solar harvesting with energy storage, maximum power point tracking, and intelligent power management. ([ScienceDirect](#)) Sourced on 05-01-2026.

For underwater robots, environmental energy harvesting is being investigated as a means of extending deployment duration. Current research includes wave energy, thermal gradients, currents, and hybrid harvesting systems. ([MDPI](#)) Sourced on 05-01-2026.

These developments establish an energy system progression: **Primary energy source → energy conversion → energy storage → power management → actuation → motion → energy recovery/harvesting → storage and reuse**.

This progression provides the foundation for considering energy harvesting as part of the overall robotic architecture rather than as an isolated component.

1.10 Future Trends in Robotic Energy Systems

One direction for future trends is the development of **hybrid energy systems**, in which more than one energy source or storage technology is combined. This can include battery-supercapacitor systems, solar-battery systems, or combinations of environmental energy harvesting technologies.

A second direction is **intelligent energy management**. Energy harvesting systems are inherently dependent on operating conditions. Solar availability changes with illumination, wave energy varies with sea conditions, and mechanical energy depends on robot motion. Consequently, future systems require energy management methods capable of coordinating energy generation, storage, and consumption.

Research on unmanned ocean vehicles has identified hybrid multisource energy harvesting, AI enabled power management, miniaturization, hydrodynamic optimization, energy storage, and adaptive control as future research directions. ([ScienceDirect](#)) Sourced on 06-01-2026.

Similarly, recent UAV research is connecting photovoltaic harvesting, storage, power electronics, aerodynamic design, and autonomous energy control rather than treating these systems independently. ([ScienceDirect](#)) Sourced on 06-01-2026.

For humanoid robots, increasing energy demand has created continued research interest in higher-energy-density batteries and improved energy-storage architectures. ([PubMed](#)) Sourced on 06-01-2026.

These trends indicate a movement toward **integrated robotic energy architectures**, in which energy generation, storage, actuation, mechanical design, and control are considered together.

1.11 Challenges Associated with Robotic Energy Harvesting

1.11.1 Energy conversion efficiency

Energy is lost during mechanical to electrical conversion, electrical conditioning, storage, and subsequent conversion back to mechanical work. Therefore, the energy recovered by a harvesting system must be evaluated against the losses introduced by the system.

1.11.2 Added mass

An energy harvesting mechanism adds components to the robot. Additional mass can increase the energy required for locomotion, particularly in humanoid and aerial robots.

1.11.3 Mechanical resistance

A harvesting mechanism connected to a moving component can introduce resistance or damping. If this resistance requires the primary actuator to consume additional energy, the overall benefit may be reduced. This makes it necessary to distinguish between: **gross harvested energy** and **net recovered energy**.

1.11.4 Intermittent energy availability

Environmental energy sources are not constant. Solar radiation varies with time and weather, waves and currents vary with ocean conditions, and motion-based energy depends on

the robot's operating cycle. Energy storage and power-management systems are therefore necessary to compensate for fluctuations. ([The Advanced Portfolio](#)) Sourced on 06-01-2026.

1.11.5 Size and integration

The harvesting mechanism must fit within the robot without interfering with its joints, sensors, actuators, payload, or other components.

1.11.6 Environmental durability

Underwater systems must operate under pressure, corrosion, biofouling, and changing temperature conditions. Wave energy research identifies corrosion, biofouling, variable ocean conditions, low energy density, and scalability as continuing challenges. ([ScienceDirect](#)) Sourced on 06-01-2026.

Aerial systems must deal with changing solar irradiance, weather, temperature, aerodynamic constraints, and structural mass. ([ScienceDirect](#)) Sourced on 06-01-2026.

1.13 Aim of the Research

The aim of this research is to investigate the feasibility of **energy harvesting from motion in humanoid and general robotic systems**, with emphasis on identifying harvestable forms of mechanical energy, evaluating the amount of energy that can be recovered, and investigating suitable mechanisms for integrating energy harvesting into robotic architectures.

1.14 Research Objectives

To achieve this aim, the research will pursue the following objectives:

- To identify the types of mechanical motion and environmental interactions in humanoid and general robotic systems that can provide harvestable energy.
- To quantify the amount of energy that can potentially be recovered from selected robotic movements and determine the effect of energy harvesting on the robot's movement and performance.
- To evaluate electromagnetic, piezoelectric, triboelectric, mechanical, and other relevant energy-harvesting mechanisms according to their energy output, efficiency, mass, integration requirements, and suitability for robotic applications.
- To investigate appropriate energy-storage and power-management approaches for storing and utilizing the energy recovered from robotic motion.

- To develop a conceptual/CAD-based integration of a selected energy-harvesting mechanism into a robotic system and evaluate its operation through simulation and robotic software environments.

1.15 Research Questions

The objectives lead to the following research questions:

- What types of mechanical motion and environmental interaction in humanoid and general robotic systems generate recoverable energy?
- How much energy can be recovered from selected robotic movements without significantly affecting the robot's movement, stability, or performance?
- Which energy-harvesting mechanisms are suitable for converting the identified mechanical energy into usable electrical energy?
- How can recovered energy be stored and managed for subsequent use within the robotic system?
- How can an energy-harvesting mechanism be integrated into the mechanical and control architecture of a robotic system?

1.17 Significance of the Research

This project is inspired by the broader goal of advancing energy systems and actuation technologies for next generation humanoid and general robots. By contributing to the understanding of motion based energy recovery, this research seeks to support the development of more autonomous, energy efficient, and sustainable robotics platforms. Ultimately, the outcome of this study could serve as a stepping stone for further innovation in self sustaining robotics, especially in low resource and off grid environments where power availability is limited like in Nigeria and Africa at Large.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

Energy consumption is one of the major limitations affecting the autonomy and operating time of modern robotic systems. Humanoid and mobile robots require energy for locomotion, actuation, sensing, computation, communication, and interaction with their environment. Most robotic systems rely on batteries as their primary energy source, but battery capacity places limitations on operating time, payload, and overall system autonomy. This has encouraged research into alternative approaches for improving robotic energy availability, including energy harvesting from the robot's own motion and from interactions with its surrounding environment [1], [2].

Energy harvesting involves converting normally dissipated mechanical or environmental energy into usable electrical energy. In robotic systems, potential sources include joint motion, vibration, braking and deceleration, foot-ground interaction, structural deformation, impacts, and other mechanical movements. Different energy-harvesting technologies have been investigated, including electromagnetic, piezoelectric, triboelectric, and mechanical systems. However, the usefulness of an energy-harvesting mechanism depends not only on the amount of energy it can generate but also on its effect on robot motion, mass, efficiency, mechanical integration, energy storage, and power management. This chapter therefore reviews available energy sources within robotic systems, energy recovery mechanisms, storage and power-management approaches, and methods for integrating energy harvesting into robotic architectures.

2.2 Harvestable Mechanical Energy in Robotic Systems

Robotic systems undergo continuous mechanical movement during locomotion and interaction with their environment. These movements represent potential sources of energy that may otherwise be lost through mechanical losses, braking, vibration, deformation, or impact. In humanoid robots, particularly relevant sources include joint motion, walking, foot-ground interaction, and the movement of the robot's limbs. Mobile robots can also produce recoverable energy during acceleration, deceleration, vibration, and interaction with uneven surfaces.

Regenerative braking provides one of the clearest examples of mechanical energy recovery. During deceleration, the kinetic energy of a moving system can be converted into electrical energy through the motor-generator system rather than being entirely dissipated. Similar principles can be applied to robotic joints, where motors can operate as generators during appropriate phases of movement [3].

Walking robots provide another important opportunity because locomotion involves repeated acceleration, deceleration, impact, and changes in mechanical energy. Research into

energy efficient legged locomotion has shown that the mechanical behaviour of the legs and joints strongly influences the energy required for walking [4]. Some researchers have therefore investigated whether energy normally lost during locomotion can be recovered and reused.

Environmental interaction provides additional possibilities. Vibrations, structural deformation, contact forces, and impacts can potentially be converted into electrical energy using piezoelectric or triboelectric mechanisms [5], [6]. The available energy will, however, depend strongly on the movement characteristics, frequency, force, displacement, and operating conditions of the robot.

Therefore, identifying harvestable energy sources requires an understanding of the robot's mechanical motion rather than simply selecting an energy-harvesting device. The characteristics of each movement determine the type and amount of energy that can realistically be recovered.

2.3 Energy Recovery from Robotic Movement

The amount of energy available from robotic motion depends on the mechanical behaviour of the system. For rotational motion, recoverable mechanical energy can be associated with joint torque and angular displacement, while translational motion can be considered through force and displacement. During regenerative operation, part of the mechanical energy associated with movement can be converted into electrical energy.

For a rotational robotic joint, the mechanical work can generally be expressed as:

$$W = \int \tau(\theta) d\theta$$

where W represents mechanical work, τ is joint torque, and θ is angular displacement.

The theoretical recoverable energy is not equal to the actual electrical energy obtained because losses occur within the mechanical transmission, motor-generator, power electronics, and storage system. A simplified relationship can therefore be expressed as:

$$E_{\text{recovered}} = \eta E_{\text{available}}$$

where $E_{\text{available}}$ is the available mechanical energy and η represents the overall conversion efficiency.

Research on regenerative actuation demonstrates that motors can recover energy during negative mechanical work, particularly during deceleration or controlled lowering of loads [3]. However, recovering energy can introduce additional resistance or torque into the mechanical system. This means that increasing energy recovery without considering the effect on locomotion could alter the robot's movement.

For this reason, the energy harvesting process should be evaluated using two related quantities: **energy recovered** and **effect on robotic performance**. A harvesting mechanism that produces significant energy but substantially interferes with natural movement may not be suitable for a robotic application. The proposed research will therefore consider both the recovered energy and the resulting changes in movement, power consumption, and mechanical performance.

2.4 Energy Harvesting Mechanisms for Robotic Applications

Several energy harvesting mechanisms can be considered for robotic systems. The most relevant approaches include electromagnetic, piezoelectric, triboelectric, and mechanically driven systems.

2.4.1 Electromagnetic Energy Harvesting

Electromagnetic energy harvesting uses relative motion between a magnetic field and electrical conductors to generate electricity. In robotic systems, this principle can be integrated with motors, joints, rotary mechanisms, or linear movements. Regenerative motor operation is particularly relevant because the same actuator used to produce motion can operate as a generator during appropriate phases of movement [3].

One advantage of electromagnetic harvesting is that it can provide relatively useful electrical power when sufficient mechanical motion is available. However, its integration can require additional electrical and mechanical components, and the harvesting process may introduce resistance into the motion.

2.4.2 Piezoelectric Energy Harvesting

Piezoelectric materials generate electrical charge when mechanically deformed. They are therefore suitable for harvesting energy from vibration, strain, pressure, and repeated mechanical deformation. Piezoelectric harvesters have been widely studied for small scale energy harvesting because of their relatively compact form and ability to operate from mechanical vibrations [5].

For robotics, piezoelectric devices may be integrated into joints, flexible structures, feet, or other components that experience repeated deformation. Their main limitation is that the available power is generally dependent on the magnitude and frequency of mechanical excitation.

2.4.3 Triboelectric Energy Harvesting

Triboelectric nanogenerators generate electrical energy through contact electrification and electrostatic induction between different materials. They have attracted interest because they can

operate under different types of mechanical motion, including contact, separation, sliding, and vibration [6].

The flexibility of triboelectric systems provides opportunities for integration into robotic surfaces, joints, wearable robotic systems, and mechanical interfaces. However, factors such as material durability, output stability, environmental conditions, and integration into moving robotic structures must be considered.

2.4.4 Mechanical Energy-Harvesting Systems

Mechanical harvesting approaches can use mechanisms such as springs, gears, ratchets, compliant structures, or other mechanical energy storage and conversion arrangements. Mechanical elements can temporarily store energy from movement and subsequently release it through a generator or other conversion mechanism.

Such mechanisms may be useful in robotic joints because they can interact directly with the robot's mechanical movement. However, additional mechanical components increase system complexity and mass. The mechanism must therefore be designed so that its benefits outweigh the additional mechanical load.

2.5 Comparison of Energy Harvesting Mechanisms

The selection of an energy harvesting mechanism cannot be based only on its theoretical electrical output. A mechanism that produces high energy but adds significant mass or interferes with robotic movement may be unsuitable for practical integration.

Important comparison parameters include energy output, conversion efficiency, mass, size, operating frequency, mechanical loading, durability, integration complexity, and power conditioning requirements.

Mechanism	Energy Source	Advantages	Limitations / Considerations
Electromagnetic	Rotation and translation	Suitable for relatively large mechanical movements; can provide useful electrical output	Mechanical integration, generator size, and possible resistance

Piezoelectric	Vibration, strain, and deformation	Compact, lightweight, and suitable for repeated mechanical excitation	Relatively low power output and dependence on vibration characteristics
Triboelectric	Contact, separation, and sliding	Flexible and suitable for different forms of mechanical motion	Material durability, environmental sensitivity, and output stability
Mechanical	Joint and structural movement	Direct integration with the robot's mechanical system	Additional mass, mechanical complexity, and possible movement resistance

Table 2.1

The appropriate technology therefore depends on the type of robotic movement being harvested. Large rotational movements may favour electromagnetic approaches, while vibration and small deformation may be better suited to piezoelectric or triboelectric mechanisms. This comparison provides the basis for selecting candidate technologies for the conceptual design stage of the research.

2.6 Effect of Energy Harvesting on Robotic Movement and Performance

An important consideration in robotic energy harvesting is that the process of extracting energy can influence the original movement. Energy cannot be recovered from a mechanical system without some interaction with the mechanical energy being converted. For example, regenerative braking can produce an opposing torque, while a mechanical harvesting mechanism may introduce additional resistance or inertia.

For a humanoid robot, this effect is particularly important because walking depends on carefully coordinated joint movements. Changes in joint torque or damping can influence gait characteristics, stability, speed, and energy consumption. Research into energy-efficient

humanoid locomotion has therefore considered the relationship between mechanical energy, actuator behaviour, and locomotion efficiency [4].

The harvesting mechanism should ideally recover energy that would otherwise be dissipated without significantly disturbing the required movement. This creates a design trade off between **energy recovery and locomotion performance**.

Consequently, the proposed research should evaluate parameters such as joint torque, angular velocity, movement trajectory, power consumption, stability, and recovered energy before and after introducing the harvesting mechanism. This will allow the effect of energy harvesting on the robot's performance to be quantified rather than considering energy output alone.

2.7 Energy Storage for Recovered Robotic Energy

Recovered energy is only useful if it can be stored or directly supplied to an appropriate load. Energy harvested from robotic motion may be intermittent because robotic movement is not constant. Walking, joint movement, vibration, and environmental interactions can produce changing levels of energy over time.

Batteries are widely used in robotic systems because they provide relatively high energy density and can supply power over extended periods. However, repeatedly charging a battery with small intermittent energy contributions may not always be optimal. Supercapacitors provide high power density and rapid charge-discharge capability, making them useful for short term storage and fluctuating harvested energy [7].

Hybrid storage systems combining batteries and supercapacitors can therefore provide complementary characteristics. The supercapacitor can absorb short bursts of recovered energy while the battery provides longer-term energy storage. The appropriate configuration depends on the magnitude, duration, and frequency of the harvested energy.

For the proposed research, storage selection should therefore be considered together with the characteristics of the harvesting mechanism. The storage system should be capable of accepting the recovered energy and delivering it to suitable robotic loads without creating unnecessary conversion losses.

2.8 Power Management of Harvested Energy

Energy harvesting and energy storage require a power-management system to control how recovered energy is transferred, conditioned, stored, and used. The electrical output of a harvesting device may not directly match the voltage or current requirements of the robot.

Power electronics such as rectifiers, DC-DC converters, charge controllers, and energy management circuits can be used to condition the harvested energy before storage or direct use. Maximum power point tracking and adaptive power-management techniques can also be applied where the output of the harvesting mechanism varies with operating conditions [8].

For a robotic system, the power-management architecture must determine whether harvested energy should be stored, used immediately, or distributed to a particular subsystem. For example, recovered energy could potentially support sensors, embedded computing, communication systems, or other low-power components rather than directly powering the primary actuators.

This creates an important design relationship: **Energy Harvesting → Power Conditioning → Energy Storage → Power Management → Robotic Loads.**

2.9 Integration of Energy Harvesting into Robotic Architectures

Integrating an energy-harvesting mechanism into a robot requires consideration of the robot's mechanical, electrical, and control architectures. The harvesting mechanism must be physically located where useful mechanical energy is available without interfering with the robot's normal operation.

Possible integration points include joints, actuators, feet, compliant structures, suspension mechanisms, and other components that experience repeated motion or deformation. For example, an electromagnetic generator could be integrated with a joint or actuator to recover energy during regenerative motion, while piezoelectric materials could be incorporated into structures exposed to vibration or deformation.

Integration also requires consideration of mass and space. Additional components increase the robot's total mass, which can increase the energy required for locomotion and potentially offset some of the recovered energy. The mechanical interface must therefore be designed so that the energy recovered exceeds the additional energy cost associated with the harvesting system.

The control system is also important. The robot may need to determine when harvesting should occur and when the actuator should prioritise movement performance. This means that energy harvesting can be treated as part of the robot's overall energy-management strategy rather than as an independent component.

2.10 CAD Based and Simulation Approaches for Robotic Energy Harvesting

Computer-aided design and simulation provide a practical method for investigating energy harvesting before constructing a physical prototype. CAD models can be used to

determine where an energy harvesting mechanism could be physically integrated and to examine dimensions, mass, mechanical interfaces, and movement constraints.

Mechanical simulation can then be used to analyse forces, torques, displacement, vibration, and other parameters associated with robotic movement. These quantities can provide estimates of the mechanical energy available for harvesting. Electrical simulation can subsequently be used to estimate generator output, conversion efficiency, storage behaviour, and power management requirements.

The integration of mechanical and robotic simulation is particularly useful because the harvesting mechanism can be evaluated while the robot performs a representative movement. A conceptual workflow could therefore be: **Robotic Motion → Mechanical Energy Analysis → Harvesting Mechanism → Electrical Conversion → Power Conditioning → Energy Storage → Robotic Load**

The simulation can then compare the original robotic system with the system containing the harvesting mechanism. Relevant outputs may include recovered energy, total energy consumption, movement characteristics, actuator torque, and changes in system efficiency.

2.11 Conceptual Integration of Energy Harvesting into a Robotic System

Based on the reviewed literature, a conceptual energy harvesting robotic system can be developed around a selected robotic movement. The first stage would identify a movement with measurable mechanical energy, such as joint rotation, walking, braking, vibration, or foot ground interaction.

The selected harvesting mechanism would then be integrated into the relevant mechanical subsystem using a CAD based design. The mechanical model would provide the movement, torque, displacement, or vibration required by the harvesting device. The generated electrical energy would pass through a power conditioning stage before being stored or supplied to an appropriate robotic load.

A conceptual architecture can be represented as: **Robotic Movement → Mechanical Energy Recovery → Energy Conversion → Power Conditioning → Energy Storage → Power Management → Robotic Subsystem**

A monitoring system would measure mechanical and electrical parameters throughout the process. These measurements would allow the recovered energy and its effect on robotic performance to be evaluated.

2.12 Evaluation of Energy Harvesting Performance

The performance of an energy harvesting robotic system should be evaluated using both energy-related and robotic performance parameters. The main energy metrics can include:

- Mechanical energy available for recovery
- Electrical energy generated
- Energy conversion efficiency
- Energy recovered per movement or cycle
- Average harvested power
- Energy stored
- Energy consumed by the power-management system

The effect on the robot can be evaluated using:

- Joint torque
- Joint position and velocity
- Movement trajectory
- Locomotion stability
- Robot power consumption
- Movement duration
- Additional system mass
- Overall energy efficiency

2.13 Research Gap

Existing research has established that mechanical and environmental energy can be recovered from robotic systems using electromagnetic, piezoelectric, triboelectric, and other energy-harvesting technologies. Research has also investigated regenerative actuation, energy-efficient locomotion, vibration harvesting, and power-management techniques [3]–[8].

However, several issues remain. First, the amount of energy that can realistically be recovered depends strongly on the specific movement and operating condition of the robot. Second, energy harvesting can introduce additional mechanical resistance, mass, and complexity, meaning that the energy recovered may not always translate into a net improvement in system efficiency. Third, energy harvesting, storage, and robotic control are often investigated as separate components rather than as one integrated system.

There is therefore a need to examine the complete relationship between robotic motion, energy availability, harvesting mechanism, mechanical impact, energy storage, power management, and overall robotic performance.

The proposed research addresses this gap by identifying harvestable mechanical energy in selected robotic movements, quantifying the recoverable energy, comparing suitable harvesting

mechanisms, investigating storage and power management approaches, and developing a CAD based integration of a selected mechanism for simulation.

2.14 Summary

This literature reviewed energy harvesting from motion in humanoid and general robotic systems. The review first examined mechanical movements and environmental interactions that can provide recoverable energy, followed by methods for quantifying available and recovered energy. Electromagnetic, piezoelectric, triboelectric, and mechanical harvesting mechanisms were then examined based on their operating principles and suitability for robotic applications.

The literature review also considered the effect of energy harvesting on robotic movement, as well as energy-storage and power-management requirements. Finally, CAD and simulation approaches were discussed as methods for integrating and evaluating a selected harvesting mechanism within a robotic architecture.

The literature indicates that the feasibility of robotic energy harvesting depends on more than the electrical output of the harvesting device. The mechanical energy available, conversion efficiency, added mass, effect on movement, storage efficiency, and integration requirements must all be considered. These findings provide the basis for the methodology, where a suitable robotic movement and harvesting mechanism can be selected, modelled, integrated, simulated, and evaluated.

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CHAPTER THREE

METHODOLOGY